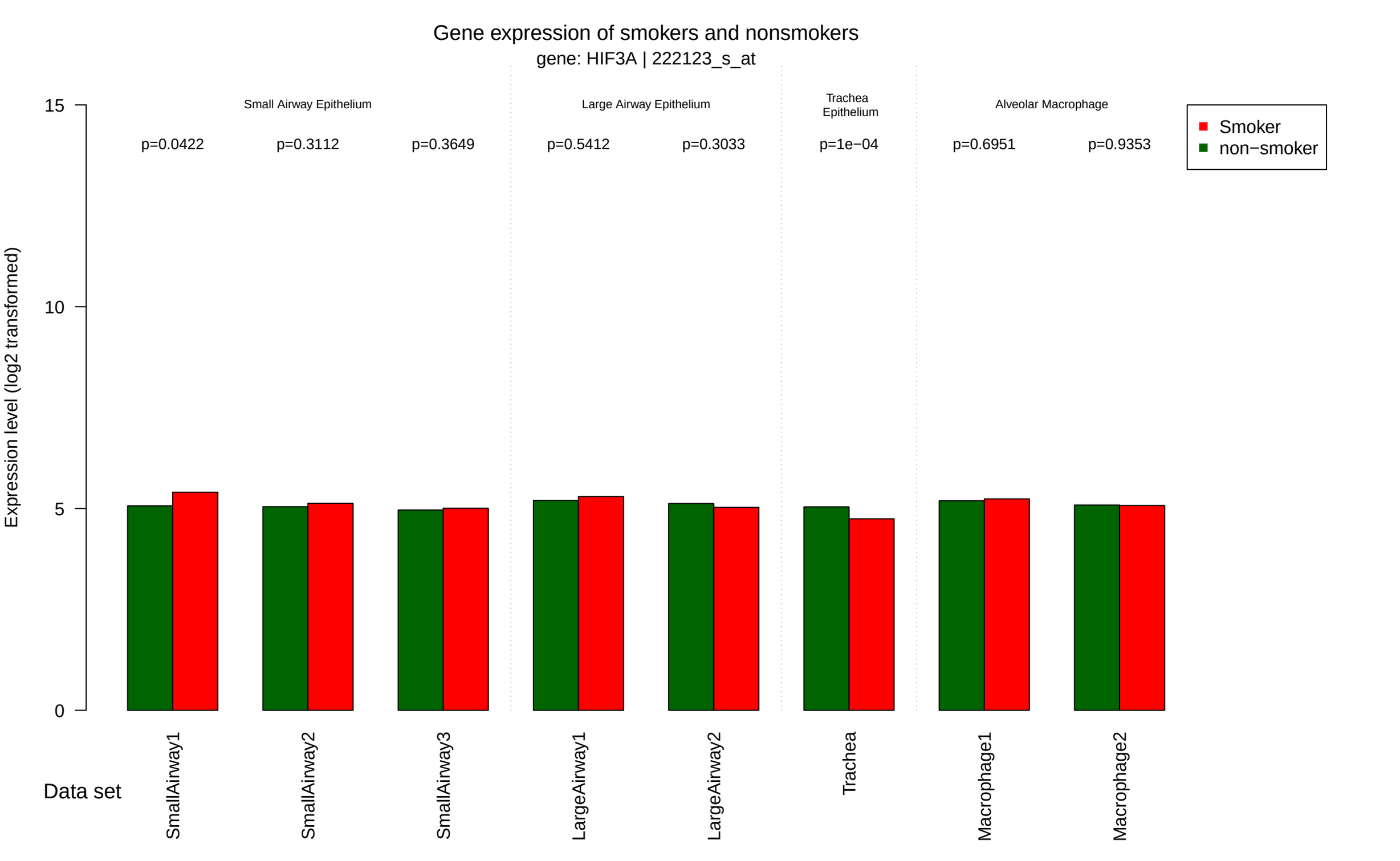
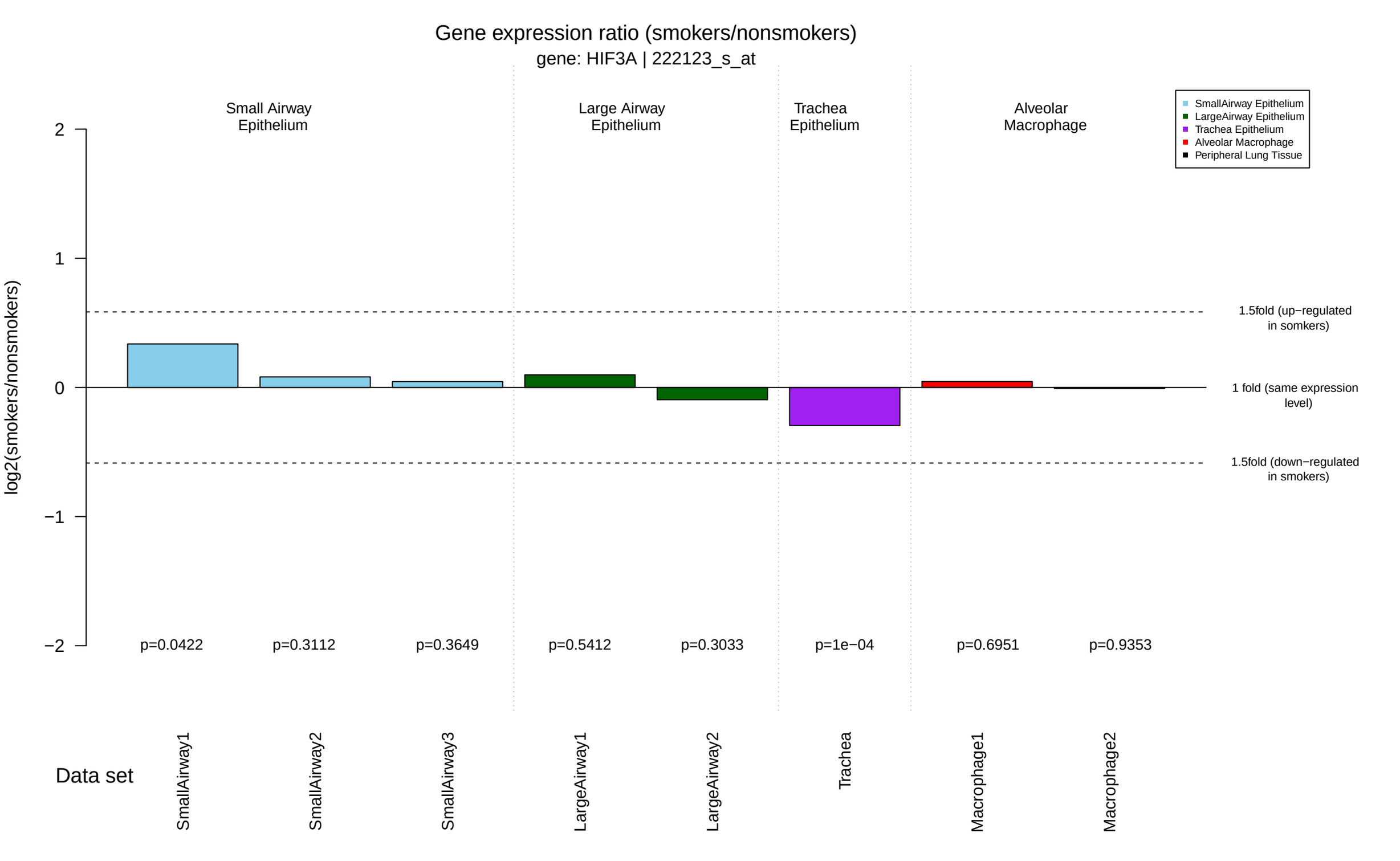
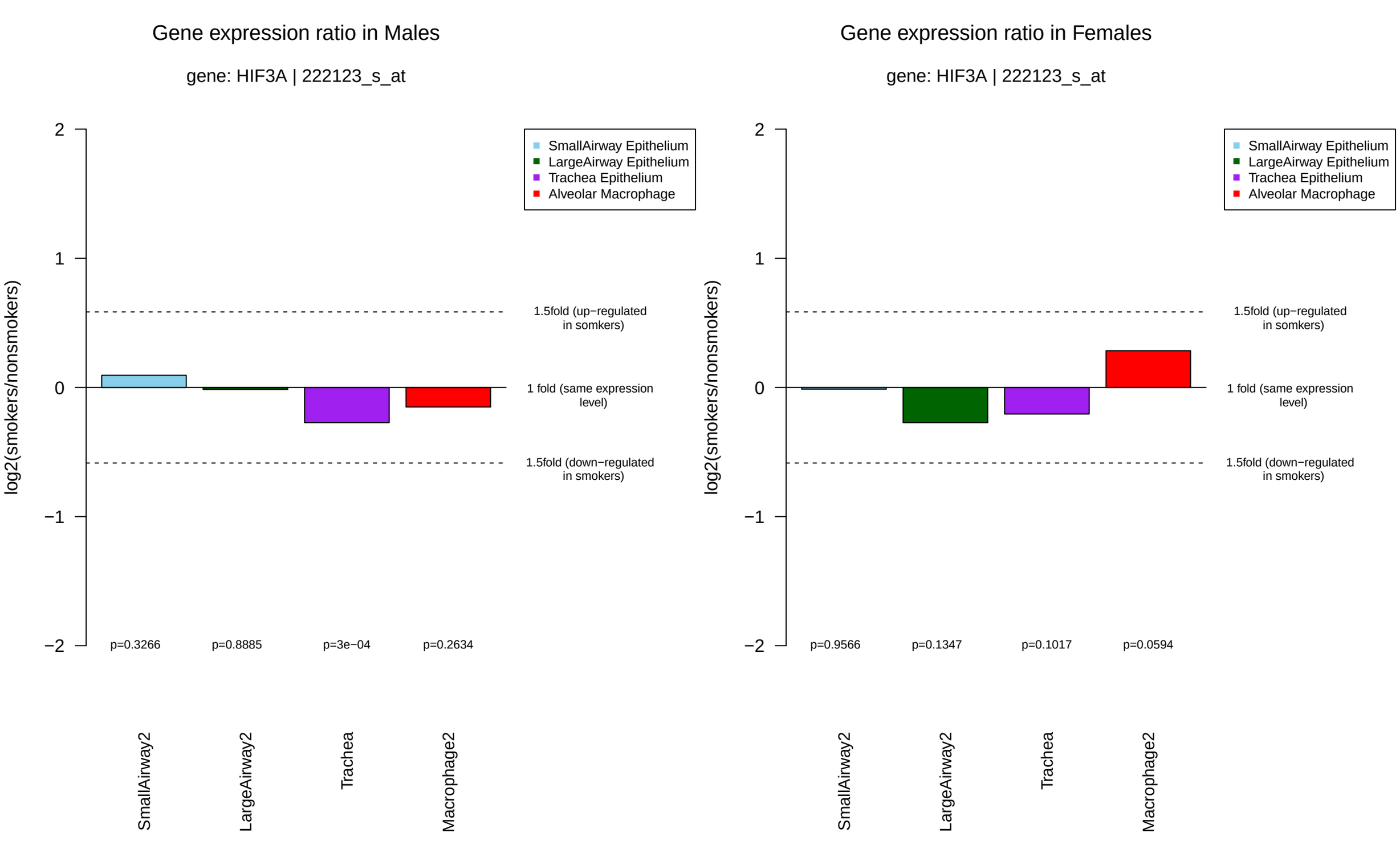
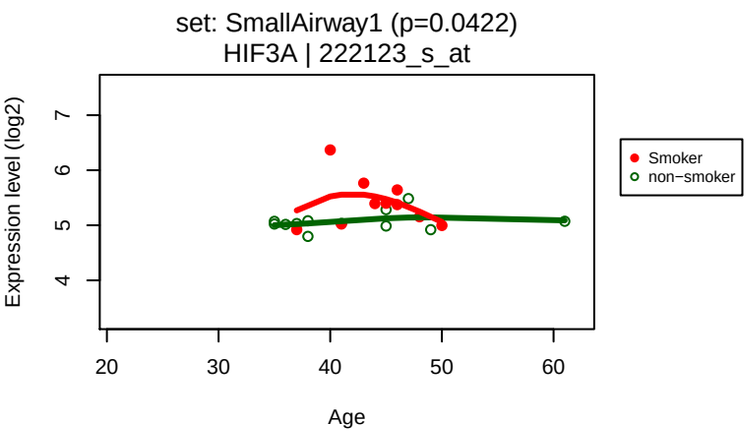
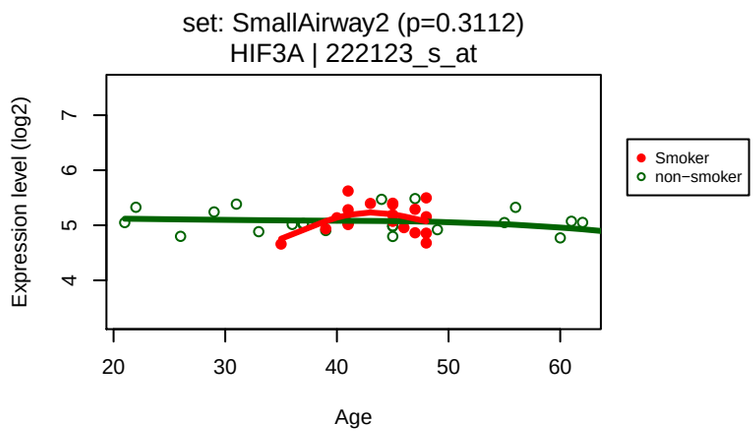
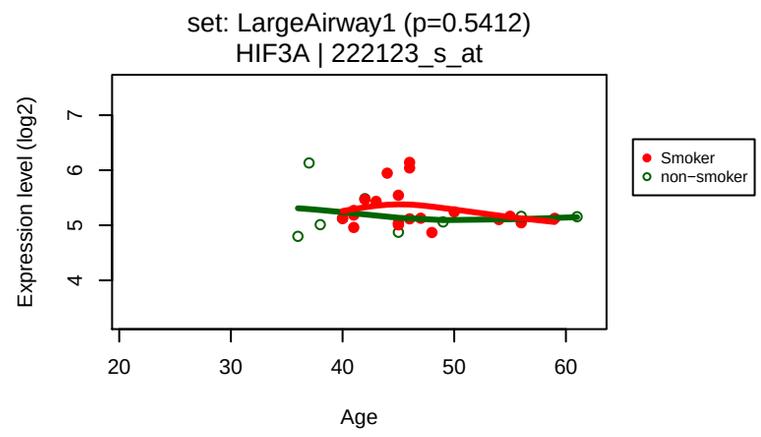

Gene expression of smokers and nonsmokers
gene: HIF3A | 222123_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0422
p=0.3112
p=0.3649
p=0.5412
p=0.3033
p=1e−04
p=0.6951
p=0.9353
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

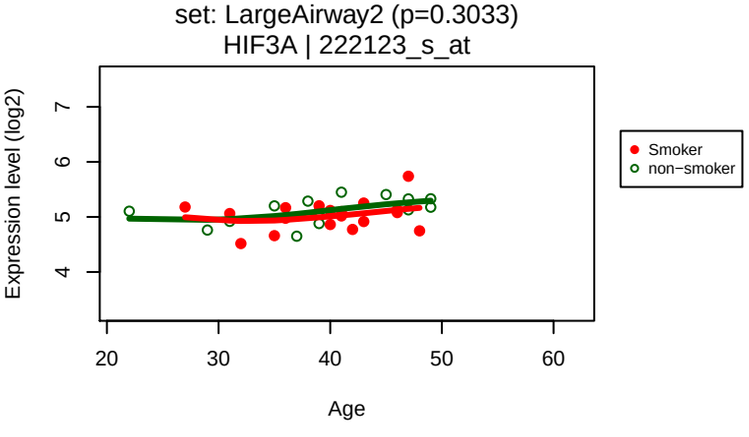

Gene expression ratio (smokers/nonsmokers)
gene: HIF3A | 222123_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0422
p=0.3112
p=0.3649
p=0.5412
p=0.3033
p=1e−04
p=0.6951
p=0.9353
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

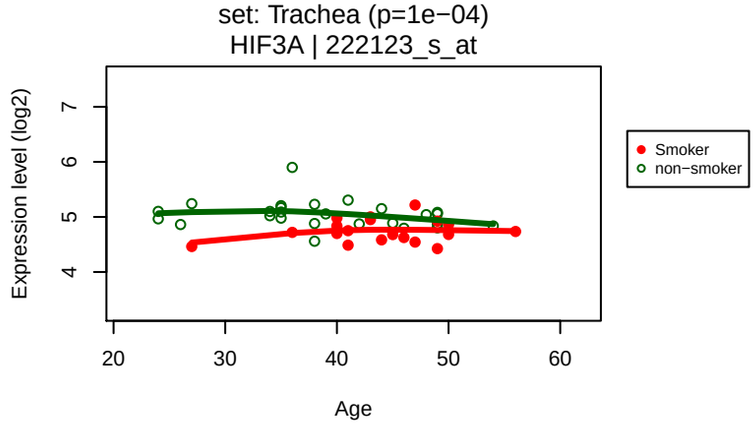

Gene expression ratio in Males
gene: HIF3A | 222123_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3266
p=0.8885
p=3e−04
p=0.2634
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIF3A | 222123_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9566
p=0.1347
p=0.1017
p=0.0594
SmallAirway2
LargeAirway2
Trachea
Macrophage2

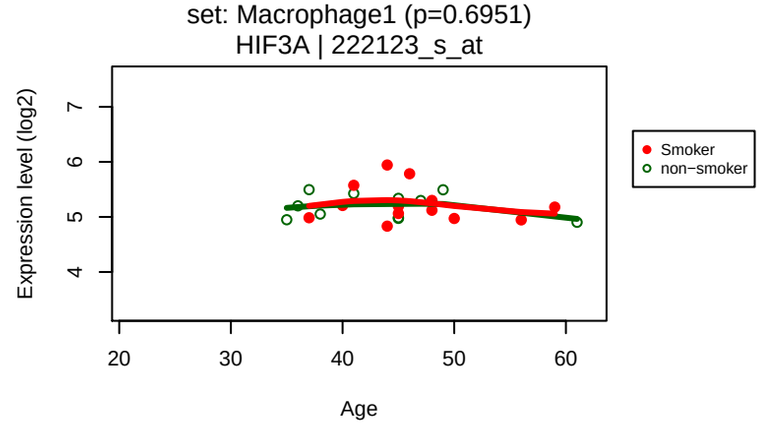

set: SmallAirway1 (p=0.0422)
HIF3A | 222123_s_at
Expression level (log2)
Age
Smoker
non−smoker

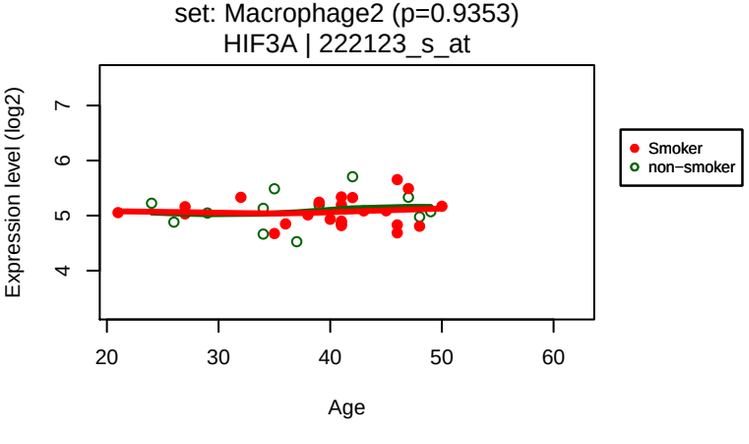

set: SmallAirway2 (p=0.3112)
HIF3A | 222123_s_at
Expression level (log2)
Age
Smoker
non−smoker

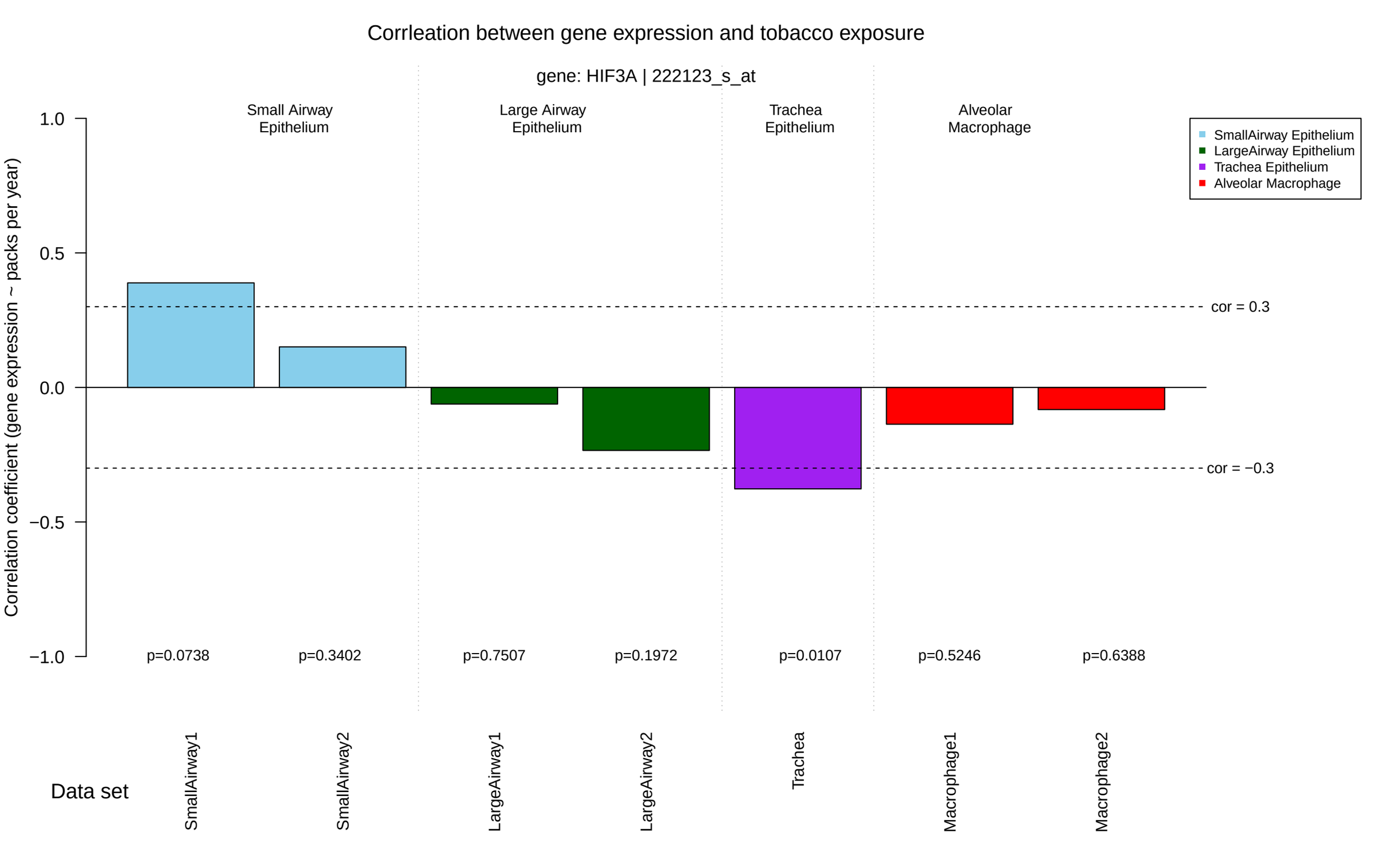

set: LargeAirway1 (p=0.5412)
HIF3A | 222123_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3033)
HIF3A | 222123_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
HIF3A | 222123_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6951)
HIF3A | 222123_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9353)
HIF3A | 222123_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIF3A | 222123_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0738
p=0.3402
p=0.7507
p=0.1972
p=0.0107
p=0.5246
p=0.6388
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2